BREAST SURGERY

MBBS IV WCS #76

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BREAST SCREENING

Breast cancer screening

- Detect early breast cancers
- Self breast examination
- Clinical breast examination
- Mammogram

Self breast examination

- Regular, formally taught, self examination (monthly interval)
- Meta-analysis and RCT from Shanghai
  - No difference T stage
  - False positive
- International guidelines no longer recommend women to perform regular self examination
- Advised to be aware of changes in their breasts

Clinical Breast Examination

- Clinical breast examination is physical examination of the breasts and axilla by trained clinician
- Three RCTs from The Philippines and India
  - Detection of smaller tumor
  - No data on improvement of survival
- Studies from the Cochrane, the American Cancer Society (ACS), and the U.S. Preventive Services Task Force (USPSTF)
  - Insufficient evidence to support clinical breast exam as screening

Mammogram screening

- The ACS systematic review
  - Approximately 20% reduction in BC mortality of average-risk women
  - When comparing with women aged <50, screening women aged ≥50 was associated with slightly greater BC mortality reduction.

- Harms
  - False positive (Overdiagnosis)
  - Overtreatment
  - False negative (False reassurance)
  - Radiation
  - Pain and discomfort

Current practice in the West

- Age based (US, Australia, Canada, UK, and EU)
  - Screening age from 50 to 69
  - Extension of screening age to 74 in US, Canada and Australia
- Mammogram every 2 years is recommended

HK Cancer Expert Working Group on Cancer Prevention and Screening

- **CEWG Recommendations on BC screening for average risk:**

Women aged 44-69

With certain combinations of personalised risk factors
(including presence of history of breast cancer among first-degree relative, a prior diagnosis of benign breast disease, nulliparity and late age of first live birth, early age of menarche, high body mass index and physical inactivity)

- Recommended to consider mammography screening every two years
- They should discuss with their doctors on the potential benefits and harms

Management of breast conditions

Clinical history for patients with breast symptoms

- Breast symptoms
  - Lump
  - Pain
  - Nipple discharge

- Further characterization
  - Duration of symptoms
  - Changes of symptoms
  - Unilateral or Bilateral
  - Characteristics

- Familial risks
  - First / second degree (Age of onset)

- Hormonal risk
  - Menarche / Menopause
  - Gestational history
  - Breast feeding history
  - Oral contraceptive pills
  - Hormonal replacement therapy

- Previous breast screening (if any)

- Consent / Chaperone / Curtain

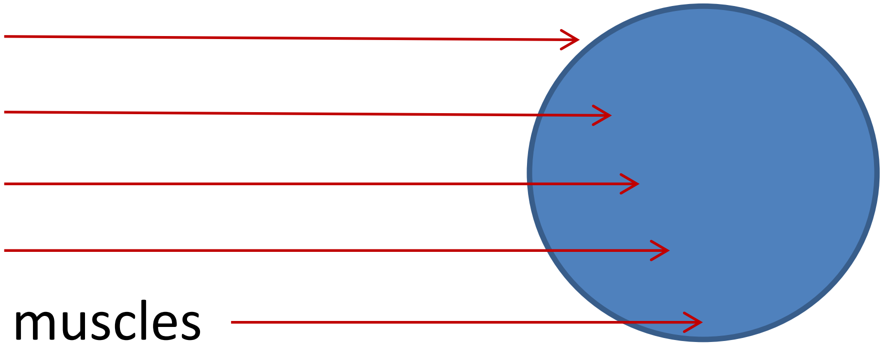
- Positioning
- Exposure

- Inspection
  - Symmetry
  - Scar
  - Skin changes
  - Special maneuver (Raise up patient’s arms)

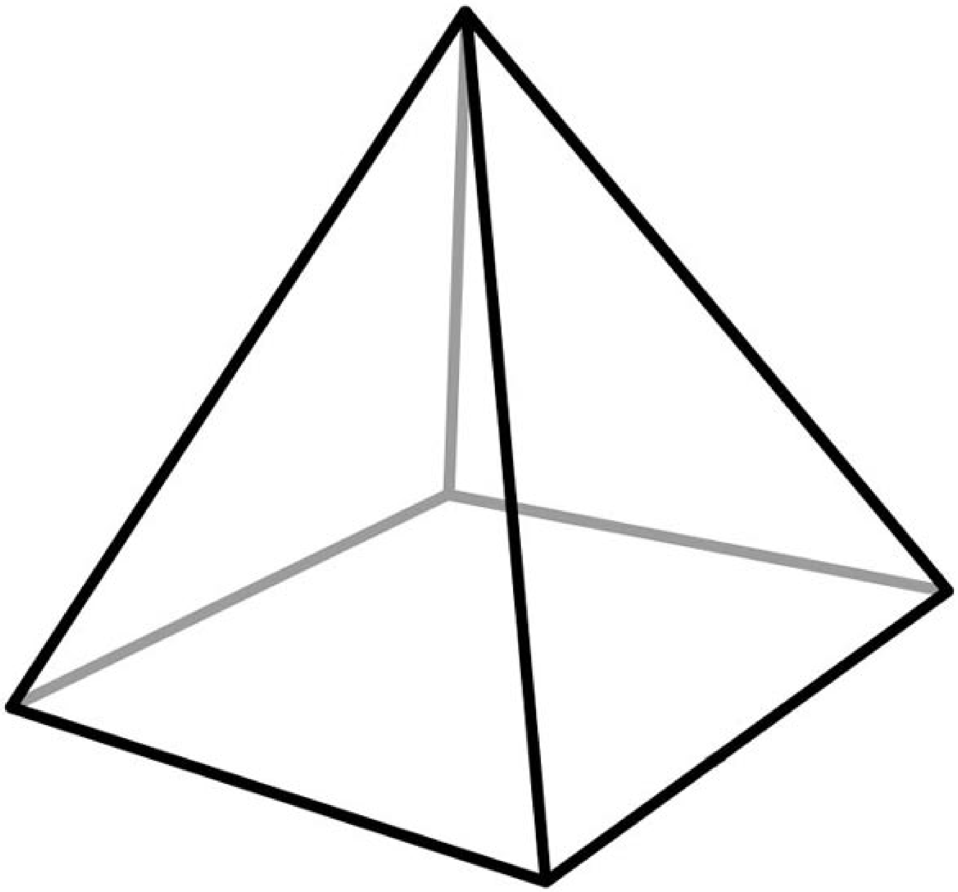
- Palpation
  - Breast mass
  - Axillary lymph nodes
  - Nipple discharge (if applicable)

Description of (breast) lumps

- Location
  - Clock face / distance from nipple
- Size
  - Precise measurements
- Characteristics
  - Border
  - Surface
  - Consistency
  - Tenderness
  - Attachment to skin / muscles



Palpation of Axillary Lymph Nodes



- Anterior
- Medial
- Posterior
- Lateral
- Apex

- **Triple Assessment**
  - Clinical (History + examination)
  - Radiological exam
  - Pathological exam

- Sensitivity 99.6% and specificity 93%

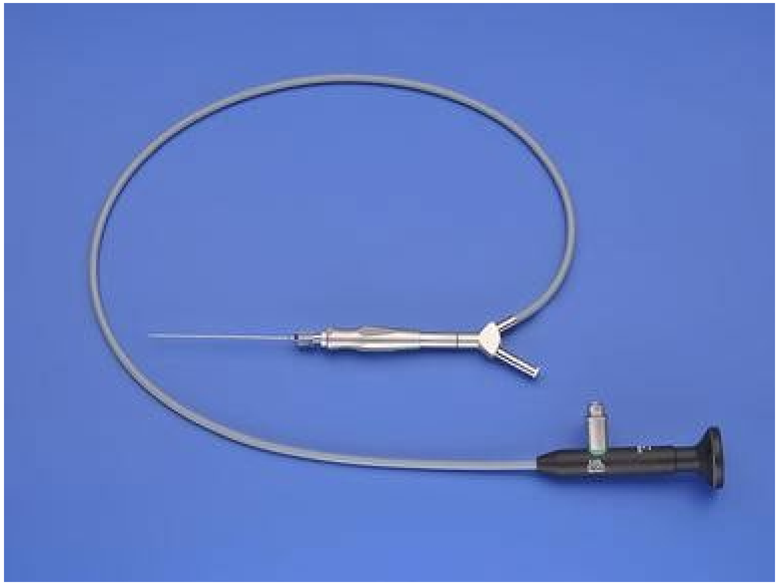
Nipple discharge

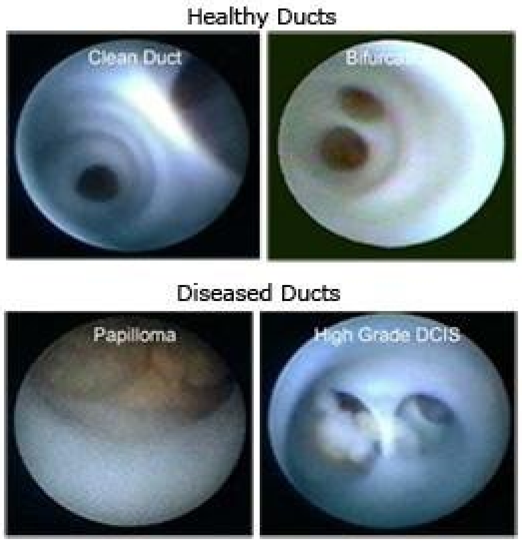
- Symptom
  - Unilateral / bilateral
  - Characteristics
    - Blood stained / milky / serous
  - Spontaneous / on manual expression
- Sign
  - Unilateral / bilateral
  - Single duct or multiple duct
  - Nature of discharge (color)
  - Any palpable breast mass / axillary lymph node

Single duct milky discharge

Ductoscopy

- Allows direct visualization of the lining of lactiferous ducts via a small fiberoptic scope





Treatment

- Microdochectomy
  - Excision of the diseased duct
  - Usually guided by ductogram / ductoscopy

DDx of breast mass

- Benign
  - Fibroadenoma
  - Cysts
  - Phyllodes tumor (Benign)
  - Others (skin lesions etc)

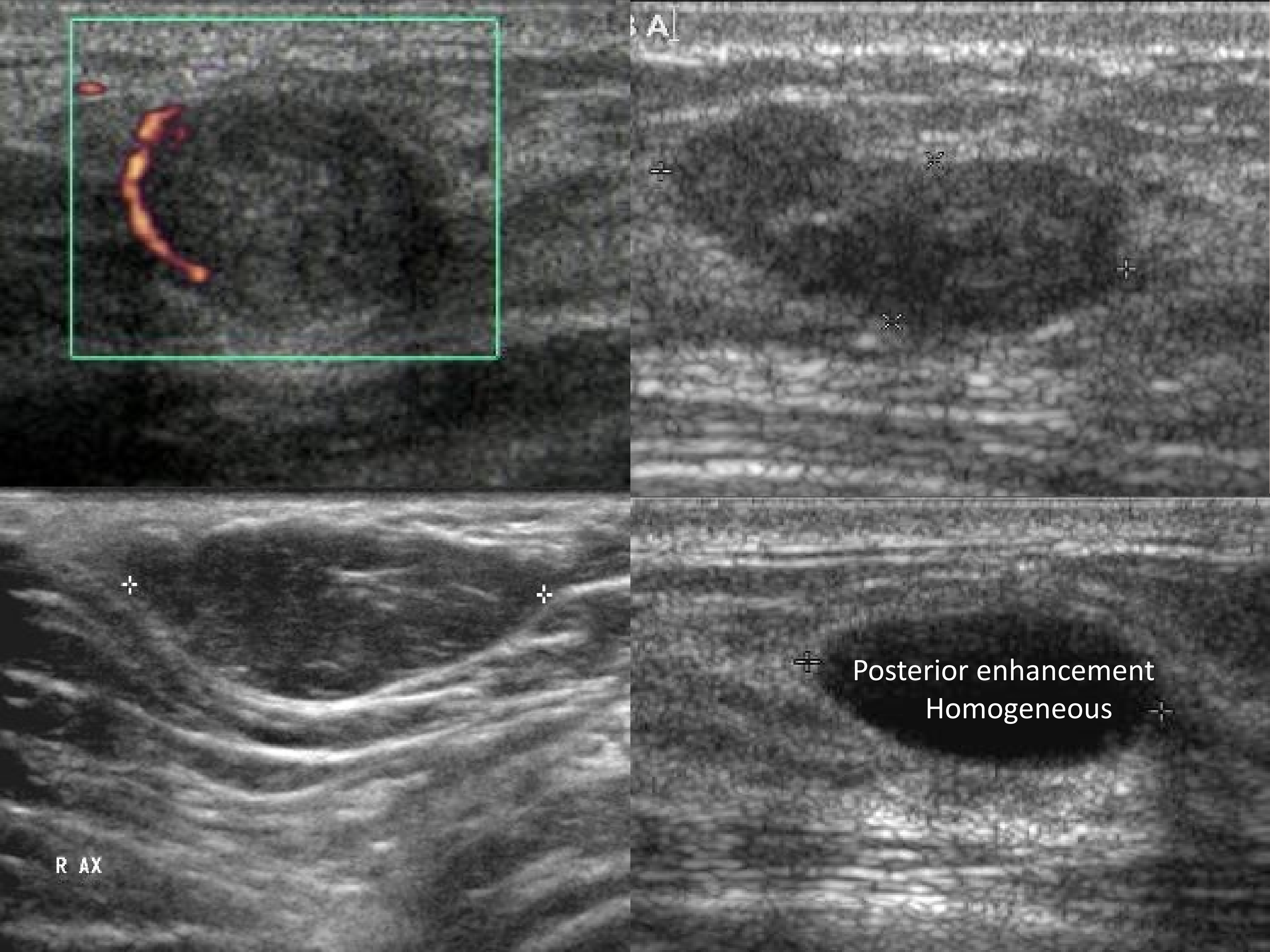
- Malignant
  - Carcinoma
    - In situ
    - Invasive
  - Phyllodes tumor (Malignant)

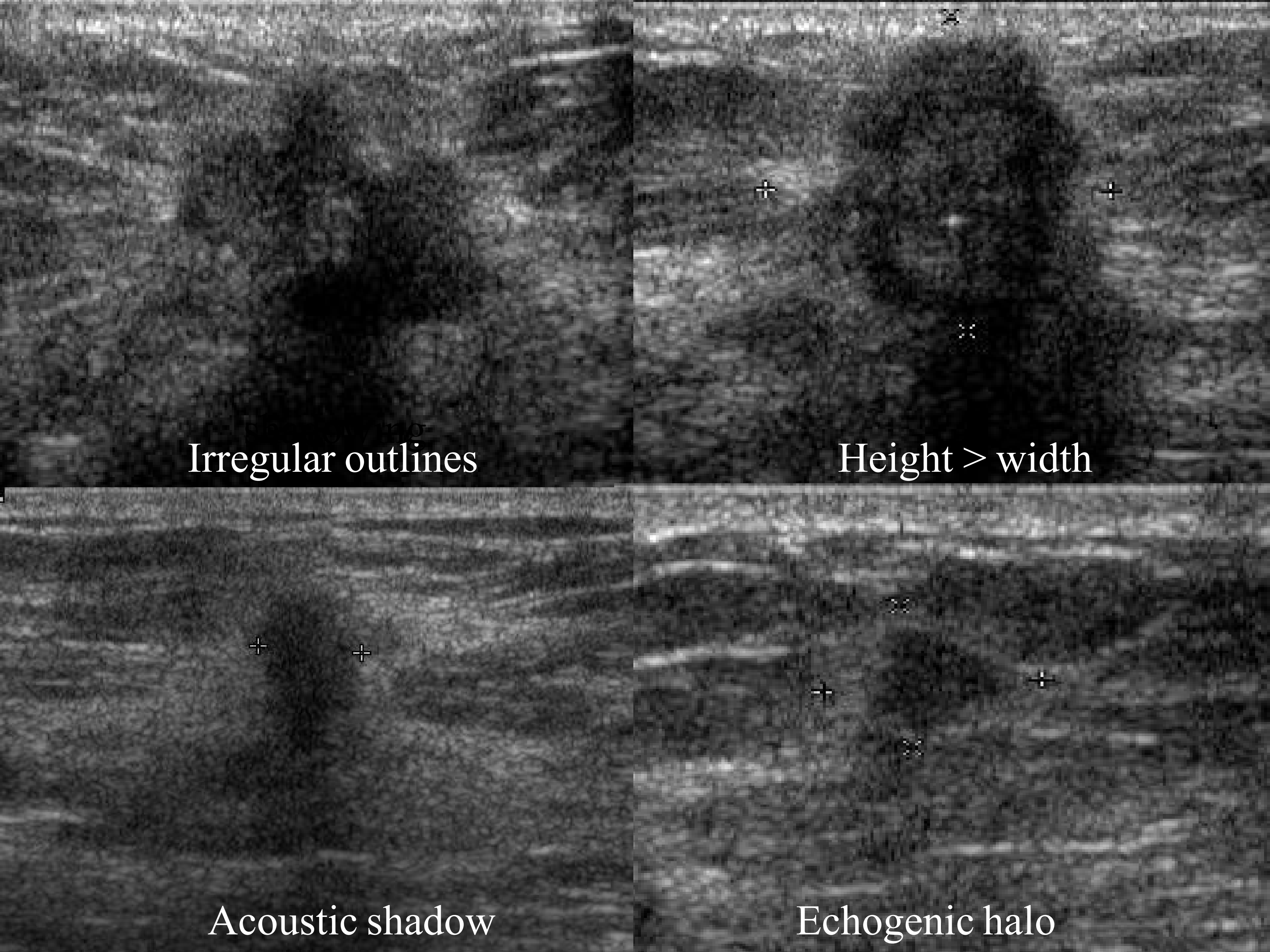
Breast imagings

- Ultrasound
  - Younger patients
  - Visualize lesions better in dense breasts
  - Characterize breast lumps

- Mammogram
  - Characterize calcifications

- MRI breasts


A
R AX
Posterior enhancement
Homogeneous

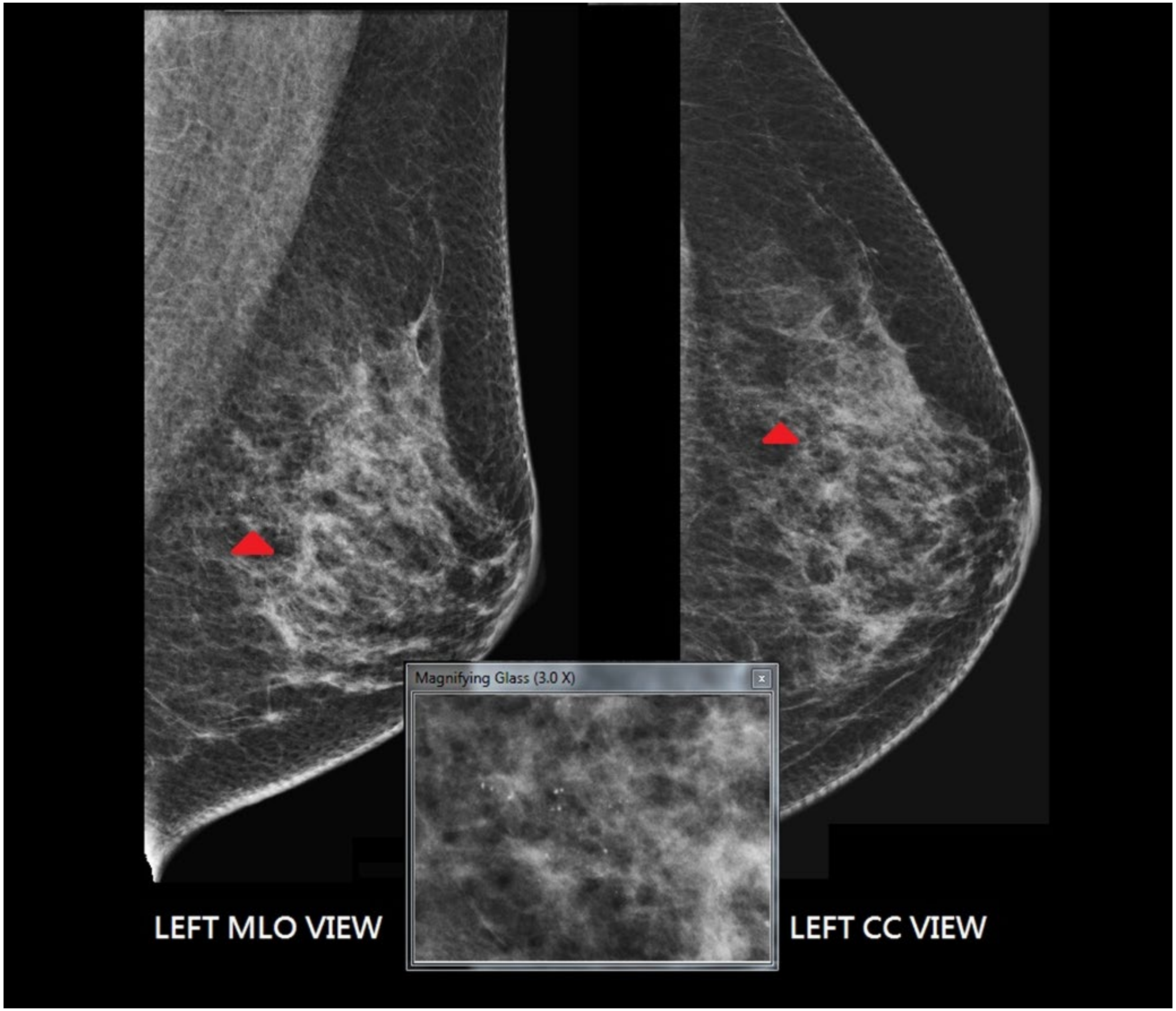


Irregular outlines
Height > width
Acoustic shadow
Echogenic halo

Mammogram Interpretation

- Preparation
  - Describe the name of patient
  - Date of MMG taken
  - Report the views taken (Usually CC, MLO views)
  - Place the two sides (Side by side)

- Describe
  - Obvious asymmetry
  - Skin thickness
  - Parenchyma density
  - Obvious mass density
  - Calcification
  - Axillary lymph nodes



Abnormal calcifications

Clusters of pleomorphic microcalcifications

Breast MRI

- Screening women at **high risk** of breast cancer
- Patients with **breast implants / augmentation**
- Evaluate **questionable suspicious lesions** seen on mammography or ultrasound
- Identify patients with **clinically occult tumour** presenting with positive axillary nodes
- Monitor result of **neoadjuvant therapy**

Biopsy techniques

- Cyotology
  - Fine needle aspiration (FNAC)

- Histopathology
  - Core needle biopsy
  - Incisional biopsy
  - Excisional biopsy

All biopsy techniques can be guided by

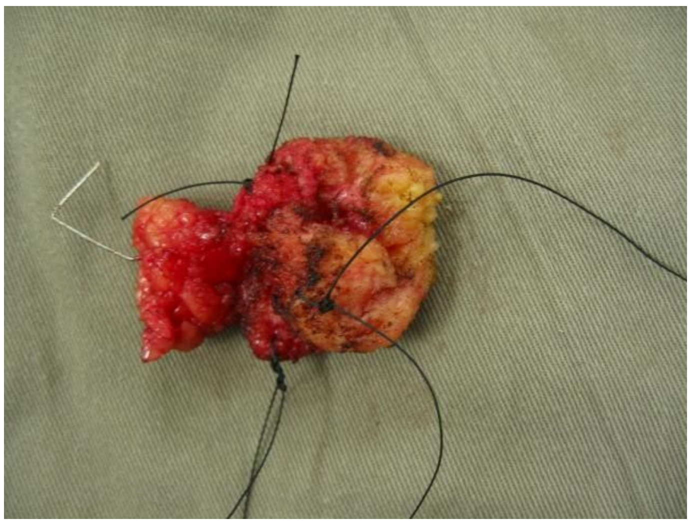
1. Free hand
2. Ultrasound
3. Mammogram (Stereotatic)
4. MRI

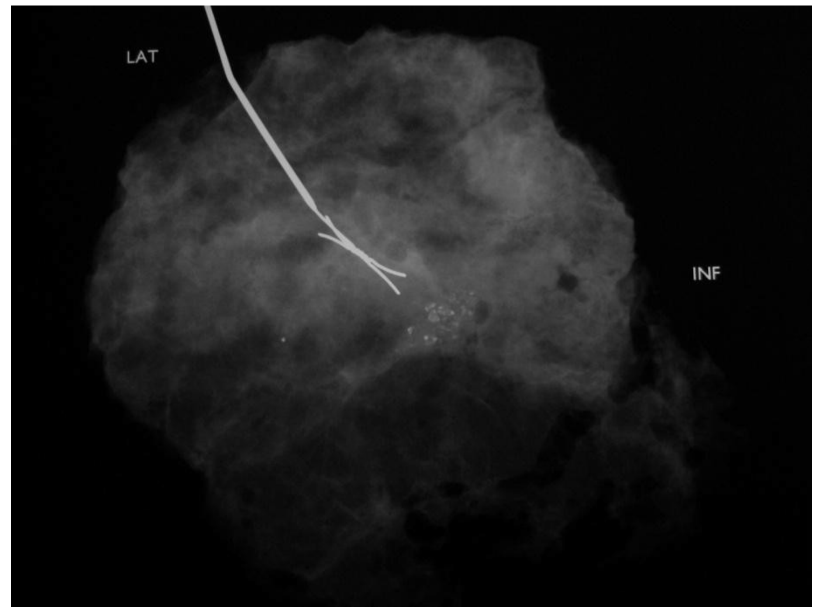
FNAC vs core biopsy

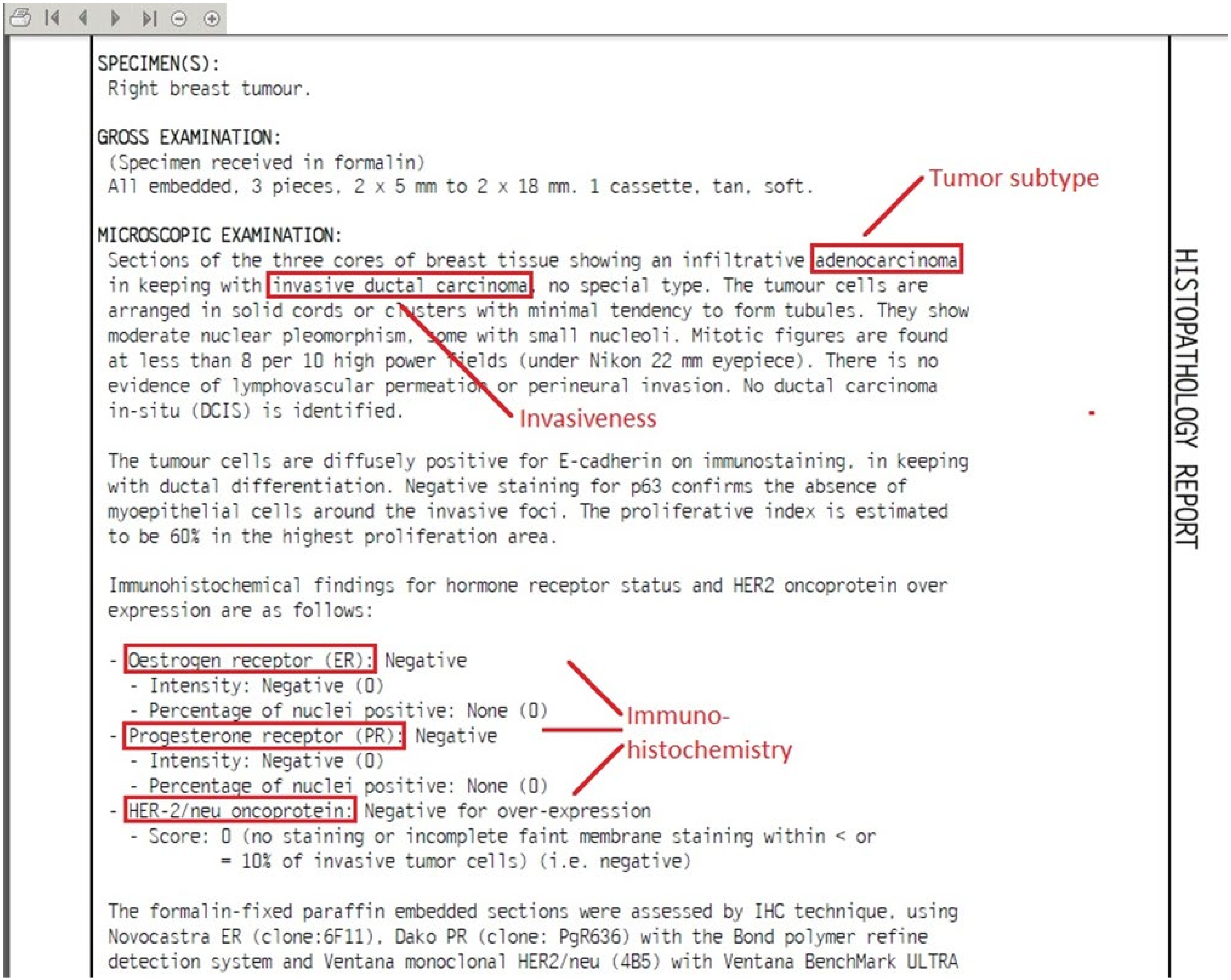
	FNAC	Core biopsy
Advantages	•Safe, simple and inexpensive •Immediately distinguishes cysts from solid masses	•Distinguishes DCIS from invasive cancer •Allows immuo-histochemical staining for ER/PR/C-erbB2 status
Disadvantages	•Cannot reliably distinguish DCIS from invasive cancer •Cannot perform immunohistochemical staining	•Relatively more invasive •Small stab incision

	Free hand	Ultrasound guided	Stereotatic (MMG)	MRI guided
Fine needle aspiration	✓	✓	Won't do FNA	Won't do FNA
Core needle biopsy	✓	✓	✓	✓
Incisional biopsy	✓	Not needed	Not needed	Not needed
Excisional biopsy	✓	✓ (USG hookwire guided)	✓ (Stereotatic hookwire guided)	✓ (MRI hookwire guided)

Example:
Stereotactic hookwire guided excisional biopsy







HISTOPATHOLOGY REPORT

SPECIMEN(S):
Right breast tumour.

GROSS EXAMINATION:
(Specimen received in formalin)
All embedded, 3 pieces, 2 x 5 mm to 2 x 18 mm, 1 cassette, tan, soft.

MICROSCOPIC EXAMINATION:
Sections of the three cores of breast tissue showing an infiltrative adenocarcinoma in keeping with invasive ductal carcinoma, no special type. The tumour cells are arranged in solid cords or clusters with minimal tendency to form tubules. They show moderate nuclear pleomorphism, some with small nucleoli. Mitotic figures are found at less than 8 per 10 high power fields (under Nikon 22 mm eyepiece). There is no evidence of lymphovascular permeation or perineural invasion. No ductal carcinoma in-situ (DCIS) is identified.

The tumour cells are diffusely positive for E-cadherin on immunostaining, in keeping with ductal differentiation. Negative staining for p63 confirms the absence of myoepithelial cells around the invasive foci. The proliferative index is estimated to be 60% in the highest proliferation area.

Immunohistochemical findings for hormone receptor status and HER2 oncoprotein over expression are as follows:

- Oestrogen receptor (ER): Negative
  - Intensity: Negative (0)
  - Percentage of nuclei positive: None (0)
- Progesterone receptor (PR): Negative
  - Intensity: Negative (0)
  - Percentage of nuclei positive: None (0)
- HER-2/neu oncoprotein: Negative for over-expression
  - Score: 0 (no staining or incomplete faint membrane staining within < or = 10% of invasive tumor cells) (i.e. negative)

The formalin-fixed paraffin embedded sections were assessed by IHC technique, using Novocastra ER (clone:6F11), Dako PR (clone: PgR636) with the Bond polymer refine detection system and Ventana monoclonal HER2/neu (4B5) with Ventana BenchMark ULTRA

Management of breast cancers

Breast cancers

- In-situ carcinoma
  - ductal carcinoma in-situ (DCIS)
  - Low, Intermediate, High Grade

- Invasive carcinoma
  - Invasive ductal (80%)
  - Invasive lobular (3%)
  - Special types
    - tubular/cribriform
    - papillary
    - mucinous
    - medullary

- Lobular carcinoma in-situ (LCIS)- a premalignant condition rather than cancer

Paget’s disease of nipple

- Eczematous changes involving the **nipple**
- Associated with malignancy within the same breast
- Malignant epithelial (Paget) cells infiltrate and proliferate in the epidermis
  - Causing thickening of the nipple and the areolar skin

Treatment of cancer (in general)

- Confirm diagnosis
- Stage the disease
- Assessment of pre-morbid condition
- Optimize nutrition
- Definitive treatment
  - Curative
  - Palliative

Surgery for breast cancers

Aims:

- *Oncological outcome*
- *Cosmetic outcome*

- **Breast surgery**
  - Mastectomy
  - Breast conserving surgery

- **Axillary surgery**
  - Sentinel lymph node biopsy (SLNB)
  - Axillary dissection (AD)

- **Reconstruction**
  - Autologous
  - Implant

Mastectomy

- Removal of entire breast (all breast tissue)

- Types
  - Simple mastectomy
    - Linear scar
  - Skin sparing mastectomy
    - Usually with reconstruction
  - Nipple sparing mastectomy (NSM)
    - Usually with reconstruction

NSM

- Selected low risk patients
- No nipple involvement
- Prophylactic mastectomy

Lumpectomy

- Removal of breast cancer with margin
  - 2mm margin for in-situ cancer
  - No cancer at the inked margin for invasive cancer

  Usually intraoperative we aim 5 – 10mm margin

- Balance between cosmetic outcome and oncologic outcome
  - Oncoplastic surgery
    - Special surgical technique

- Breast conserving treatment (BCT)
  - Lumpectomy + Radiation

Oncoplastic surgery

Oncoplastic surgery classification

Volume displacement	Examples
Level 1: < 20% breast tissue removed	Local tissue rearrangement
	Crescent mastopexy
	Doughnut mastopexy
Level 2: 20–50% of breast tissue removed	Circumvertical mastopexy design
	Reduction mammaplasty designs (including free nipple graft)
Volume replacement	Examples:
> 50% of breast tissue removed	Implant-based reconstruction
	Local/regional flap reconstruction: thoracodorsal artery perforator, etc

Definition of oncoplastic surgery: A form of breast-conservation surgery that includes oncologic resection with a partial mastectomy, ipsilateral reconstruction using volume displacement or volume replacement techniques with possible contralateral symmetry surgery when appropriate

TABLE 2 Oncoplastic surgery definition

A Consensus Definition and Classification System of Oncoplastic Surgery Developed by the American Society of Breast Surgeons

Axillary dissection

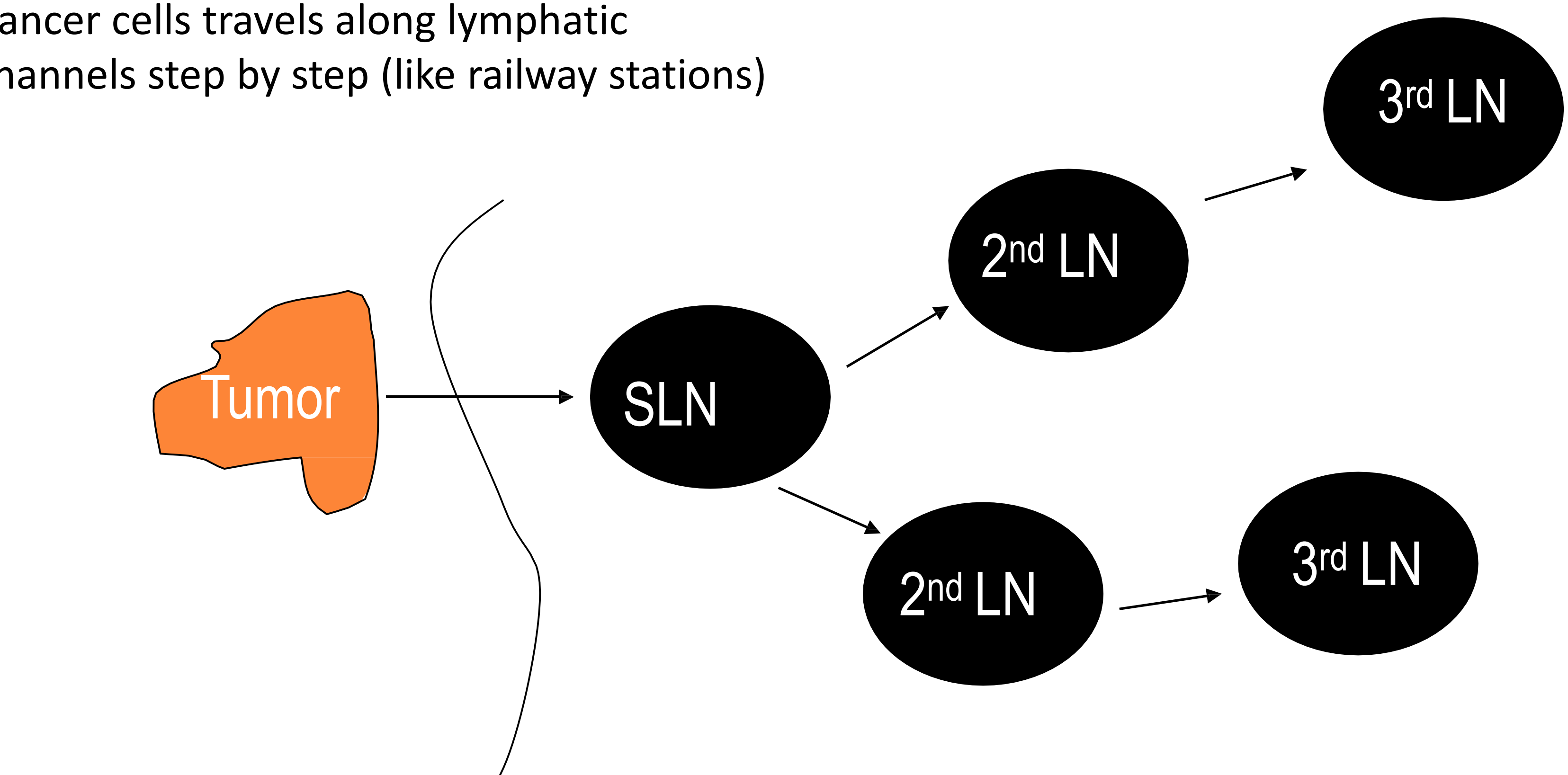
- Removal of level I & II lymph nodes
  - Pectoris minor as the landmark

- Preservation of (special caution to)
  - Long thoracic nerve
  - Thoracodorsal nerve
  - Intercostobrachial nerves

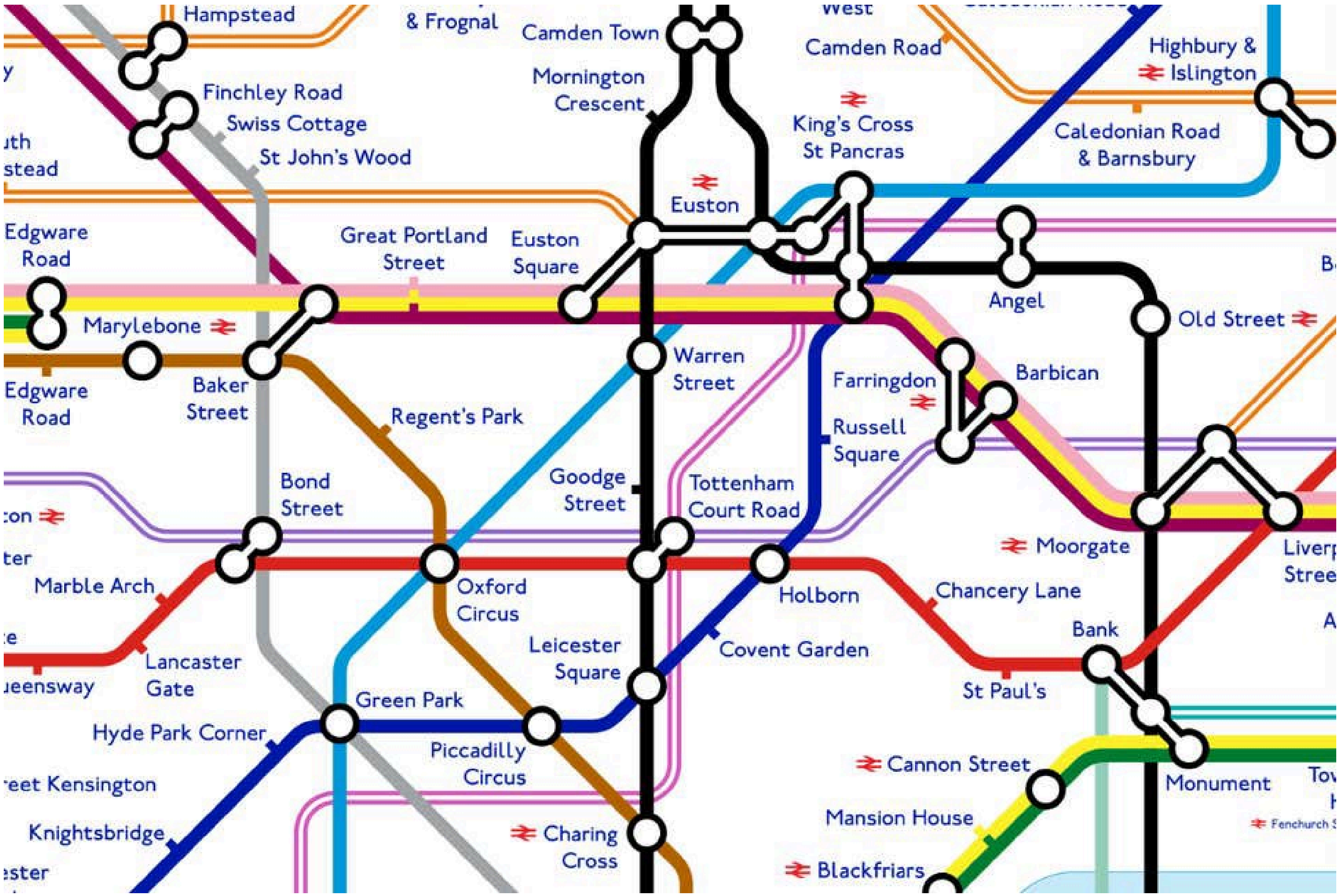
- Complications
  - Lymphedema (up to 10 – 20% risk)
  - Nerve injuries
  - General surgical complications (skin infection, scar etc)
  - Frozen shoulder

Sentinel node biopsy

Principle:
Cancer cells travels along lymphatic channels step by step (like railway stations)



How to identify the “sentinel”?



Sentinel lymph node identification

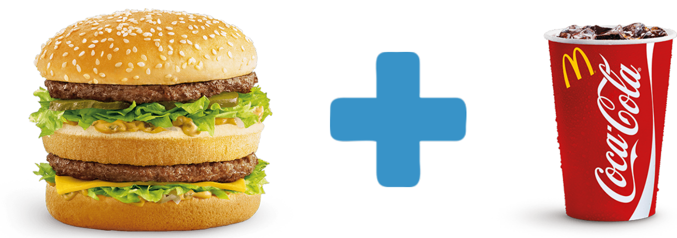
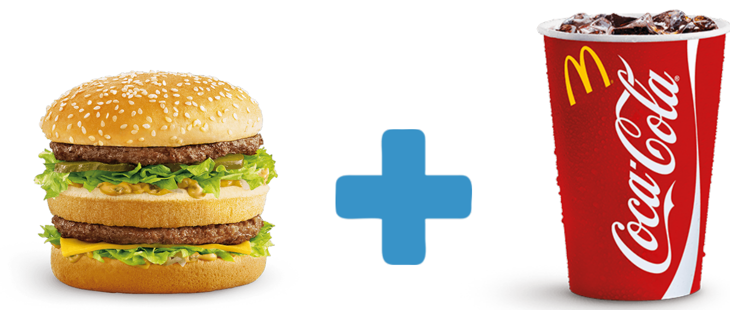
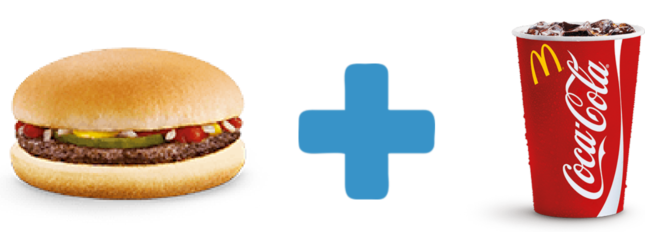
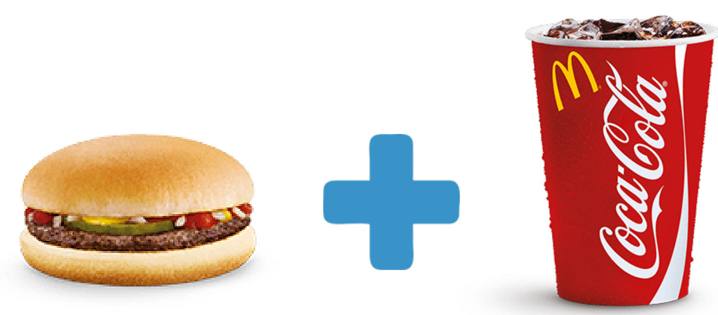
- Blue dye (patent blue)
- Radioisotope

- Others
  - Supramagnetic iron oxide (SPIO)
  - Indocyanine green (ICG)

Surgical strategy for breast cancers

Mastectomy + SLNB	Mastectomy + AD (MRM)
Breast Conserving (BCS) + SLNB	BCS + AD

Surgical strategy for breast cancers

Mastectomy + SLNB 	Mastectomy + AD (MRM) 
Breast Conserving (BCS) + SLNB 	BCS + AD 

Choice of breast surgery after NAC

- **Mastectomy or BCS?**

- Tumor response
- Breast size
- Tumor location and multiplicity
- Patient's preference

My preference matters

Reconstruction

- Implant based
- Autologous tissue based
  - Latissimus dorsi (LD)
  - Transverse rectus abdominis myocutaneous (TRAM)
- Mixture of both types

Radiotherapy

- **Locally advanced cancer**
  - Node
    - Lymph node involvement
  - Tumor
    - Large tumors
    - High grade tumors
    - Lymphovascular permeation

- **Breast conserving surgery**

Chemotherapy

- Adjuvant (or neoadjuvant)
  - Locally advanced
    - When LN are involved
    - Large tumor
  - High risk
    - Young age
    - Certain cancer subtypes (Triple negative, HER 2)
- Palliative

Hormonal therapy

- Tamoxifen vs. Aromatase inhibitor
- Adjuvant or palliative
- Its use in DCIS remains controversial